Learning Where Units End: Self-Iterated Boundary Units for Self-Supervised Event Localisation and Dense Video-Quality Rewards

[TODO: authors]

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Abstract

HuBERT-style pretraining predicts discrete units at masked positions; the representation learns where units change only in so far as the objective requires it. We measure on killer-whale field recordings that the units of a strong bioacoustic teacher change at 52% of 20 ms tokens and no more often at event boundaries than inside events, and that two label-free segmentation signals (teacher change-points, spectral flux) co-locate with event boundaries only 1.6–1.8× above chance; units built on them do not improve boundary precision. Given the oracle result, we no longer propose event-level units as a replacement. We propose *additive boundary structure*: fine 20 ms units are kept as the main target, and a second, boundary-aware target is added; the simplest instantiation is an auxiliary boundary loss, the stronger one a self-iterated coarse codebook. Self-iteration means to alternate between (a) training a student with masked unit prediction plus an onset head and (b) re-segmenting and re-clustering the corpus with the student's own boundary posterior, admitting a segmentation into the next iteration only if it passes an alignment gate measured on a small labelled set. The same machinery yields a per-frame *surprisal* that we propose as a dense, temporally localised quality reward for generated video: a boundary-aware model pretrained on keypoint and embedding trajectories of real humans scores hand and mouth artefacts as events with onsets and offsets, which no screenshot-based VLM judge provides. We lay out hypotheses, gates, experiments and risks; pilot evidence from a 14-hypothesis study on audio and telemetry is summarised.

1 Motivation from the pilot

A controlled study (companion paper) established on 17 field recordings with 613 expert events that, across HuBERT-style target sources, the onset error of matched events is flat at 56–68 ms (median) while plateaus range from 0.36 to 0.58, and that only fine-tuning moves the boundary error (to 32 ms). Three attempts to move it through the pretext task failed in an informative way:

- **Unit flicker.** Teacher-derived k-means units switch at 46% (synthetic) and 52% (field) of tokens; $P(\text{switch} \mid \text{event boundary}) = 0.48$ vs 0.52 inside events. The "where" component of the masked-prediction objective is spent on changes unrelated to events.
- **Teacher change-point segments.** Merging frames by cosine change-points of the teacher (mean 3.9 tokens, switch rate 0.20) plus a segment-start head: $F_{1\tau=.2}$ 0.412 vs 0.426, onset median 64 vs 56 ms. Segment starts fall near event boundaries with P=0.39 vs 0.25 inside.
- **Spectral-flux segments.** $P(\text{start} \mid \text{boundary})$=0.44 vs 0.24 at a 25% start rate; below the gate of 0.6 we had set, so no run was launched.
- **Oracle test (H13).** With boundaries known by construction, units averaged over true event segments beat units over random segments of the same lengths (+0.137 F_1 after FT, seed-0 bootstrap CI excludes 0) but remain below plain frame units (seed-mean gap 0.086; seed-0 bootstrap −0.055 with CI including 0), on plateau and on boundaries. Alignment is real; replacing fine units by event-level units is not.

The conclusion is not that boundaries are useless but that (i) *label-free hand-crafted segmentations are not event segmentations* and (ii) *event-level units cannot replace fine units*. Any boundary mechanism must be additive to fine units, and the first competitor to beat is the simplest one: frame units plus an auxiliary boundary loss.

2 Hypotheses

H-A (alignment drives localisation). Holding the unit vocabulary fixed, τ_{50} of a frozen probe decreases monotonically with $P(\text{unit switch} \mid \text{event boundary})$ of the target sequence. Testable with synthetic targets whose alignment is controlled.

H-B (self-iteration converges). A student's onset posterior, after one supervised calibration on $\leq$100 events, segments unlabelled audio with alignment $\geq$0.6, and units re-clustered on these segments raise $F_{1\tau\leq0.1}$ by $\geq$0.05 without lowering the plateau.

H-C (hierarchy). Fine 20 ms units for boundaries and coarse segment units for semantics in one

model (two heads, shared trunk) beat either alone once the coarse units are boundary-aligned; the failure of the mel+AVES dual codebook was a failure of the low-information codebook, not of the two-head design.

H-D (surprisal as reward). Masked-unit surprisal of a boundary-aligned model pretrained on real human motion/appearance trajectories localises artefact intervals in generated video with event-F_1 at τ=0.2 s above a forecasting-surprisal baseline (Timestral-209M) and above frame-wise VLM scores aggregated over time.

3 Method: self-iterated boundary units

Iteration 0. Pretrain as in the pilot (data2vec or km(teacher) on the Gemma-style block). Fine-tune an onset/offset head on the labelled development split only (few hundred events). Run the head over the unlabelled corpus to obtain a boundary posterior b_t.

Gate. On the labelled split compute $P(\text{start} \mid \text{boundary})$ and the start rate of the segmentation induced by thresholding b_t; accept only if alignment ≥ 0.6 at rate ≤ 0.15. If the gate fails, the iteration stops and the failure is recorded, exactly as in the pilot.

Iteration k. Segment the corpus with b_t; average student features within segments; k-means (K=100–500) on segment means; broadcast segment units to tokens; retrain with cross-entropy on units at masked positions plus binary cross-entropy on segment starts (positive weight 3). Re-estimate b_t from the new student and repeat. Stop when alignment and $F_{1\tau=.05}$ plateau.

Variants to compare. (i) hard segment units vs soft (posterior-weighted) units; (ii) with and without the fine 20 ms codebook (H-C); (iii) boundary head trained on the student's own iteration-0 units (fully unsupervised, SD-HuBERT-like) vs calibrated on $\leq$100 labelled events; (iv) multi-resolution units (MR-HuBERT style) as a baseline.

4 Application: a dense temporal judge for generated video

Problem. VLM judges score frames; FVD-style metrics score clips; artefacts on hands and teeth live in time (a finger appears for 300 ms, teeth shimmer between frames). Reinforcement-learning fine-tuning of video generators needs a reward that is localised in time and cheap.

Input series. Per frame at 25–30 Hz: hand keypoints (2×21×3, RTMPose/MediaPipe), face landmarks with emphasis on the inner mouth, DINOv2/SigLIP embeddings of hand and mouth crops, optical-flow statistics. One multichannel series; the same linear front-end as telemetry.

Pretraining. Only on real humans (VoxCeleb2/HDTF for faces, Ego4D/HO3D/InterHand for hands), boundary-aligned units from §3.

Scoring. Per-frame anomaly a_t from (a) masked-unit loss, (b) unit-transition rarity, (c) distance to the nearest centroid; two-threshold decoding into events (class: hand / mouth / teeth); event-$F_1(\tau)$ against human-annotated artefact intervals (100–200 clips, 2–3 min each to annotate). Reward for a segment is $-\bar{a}_t$; used in GRPO/DPO for Wan-2.1-class and talking-head generators available locally.

Baselines. Timestral-209M forecasting surprisal on the same series; frame-wise VLM score aggregated over time; KiMoI-style supervised kinematic detectors; VidGuard-R1 / Skyra as clip-level judges.

Control. Real videos under poor lighting and motion blur, where pose detectors also fail, to verify that surprisal responds to generation artefacts and not to difficult capture.

5 Experimental plan and gates

1. Synthetic alignment sweep for H-A on the whale mixtures (alignment controlled by construction): 1 day.
2. Iteration-0 onset head and gate on Avacha and BeamNG: 2 days. Stop if gate fails on both.
3. Two self-iterations, three seeds, frozen probe and full FT, bootstrap over recordings: 1 week.
4. H-C hierarchy ablation: 3 days.
5. Video domain: feature extraction pipeline (1 week), annotation of 150 clips (1 week, parallel), pretraining and scoring (1 week), reward integration (2 weeks). [TODO: owner]

Every run is recorded with fingerprint, code snapshot and verdict in `expdb`; a hypothesis is closed only by a verdict with run ids.

6 Risks

Self-iteration can lock in the student's own errors (confirmation); the gate and the labelled calibration split exist to catch this. Boundary posteriors on telemetry may be dominated by control discontinuities (gear shifts) rather than events of interest; per-class gates are needed. For video, pose detectors fail on exactly the frames we care about; the control set addresses this, but the signal may remain entangled with detector failure. The audio domain is limited to one species and 74 labelled minutes; EEG (Sleep-EDF, CHB-MIT) is the planned third domain for the companion paper and a fourth here.

7 Expected contribution

A pretext task whose "where" component is tied to event boundaries by construction, with an admission test that prevents the failure mode we measured; a demonstration that the resulting surprisal is a usable dense reward for video generation. If H-B fails after two iterations on two domains, the negative result, with the gate statistics, is itself the contribution: masked unit prediction cannot be made boundary-aware without labels.